

Week 3 Report

**Advanced Exploratory Data Analysis and
Feature Engineering**

House Price Prediction Dataset

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1. Introduction

This report presents a comprehensive exploratory data analysis (EDA) and feature engineering pipeline performed on the House Price Prediction dataset. The primary objective of this assignment was to understand the structure of the dataset, identify important predictive patterns, engineer meaningful features, handle skewed distributions, apply encoding and scaling techniques, and prepare the dataset for future machine learning modeling tasks.

The workflow included:

- Exploratory Data Analysis (EDA)
- Correlation analysis
- Outlier and skewness analysis
- Feature engineering
- Encoding categorical variables
- Feature scaling
- Feature selection
- Dashboard and infographic creation

The assignment demonstrates a complete preprocessing workflow commonly used in real-world machine learning projects.

2. Dataset Overview

The dataset contains residential housing information including structural characteristics, location-based features, quality ratings, and sale prices.

2.1. Dataset Characteristics

- Number of Rows: 1460
- Original Features: 92
- Final Engineered Features: 152
- Final Selected Features: 25
- Target Variable: `SalePrice`

The dataset includes:

- Numerical features
- Ordinal categorical variables
- Nominal categorical variables
- Temporal variables

3. Exploratory Data Analysis (EDA)

EDA was conducted to understand distributions, feature relationships, trends, and data quality issues.

3.1. SalePrice Distribution Analysis

The target variable `SalePrice` showed strong positive skewness, indicating the presence of high-value outliers and non-normality.

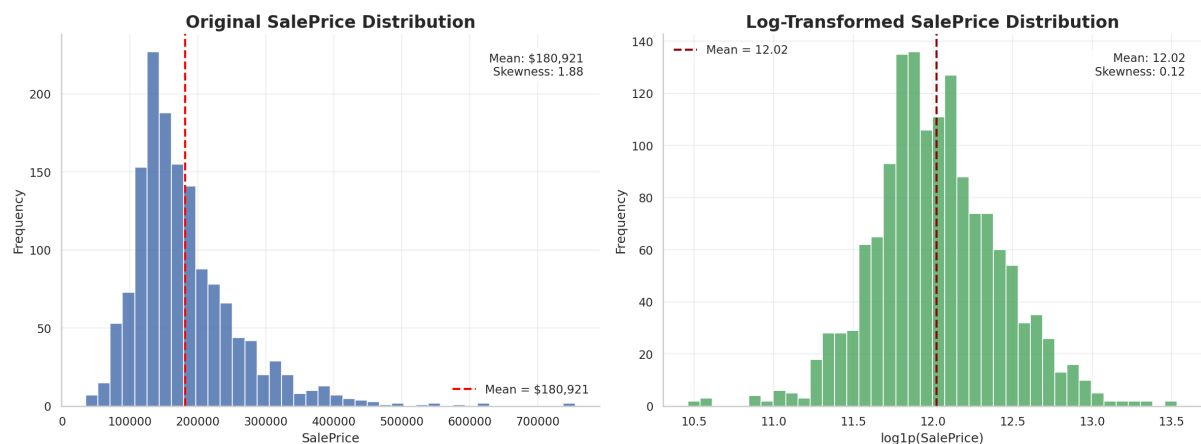


Figure 1: Original vs Log-Transformed SalePrice Distribution

3.1.1 Observations

- Original skewness was 1.88
- Distribution was highly right-skewed
- Log transformation reduced skewness to 0.12
- Transformation improved normality significantly

The transformed distribution became much more symmetric and suitable for regression-based machine learning algorithms.

3.2. Correlation Analysis

Correlation analysis was performed to identify strongly related features.

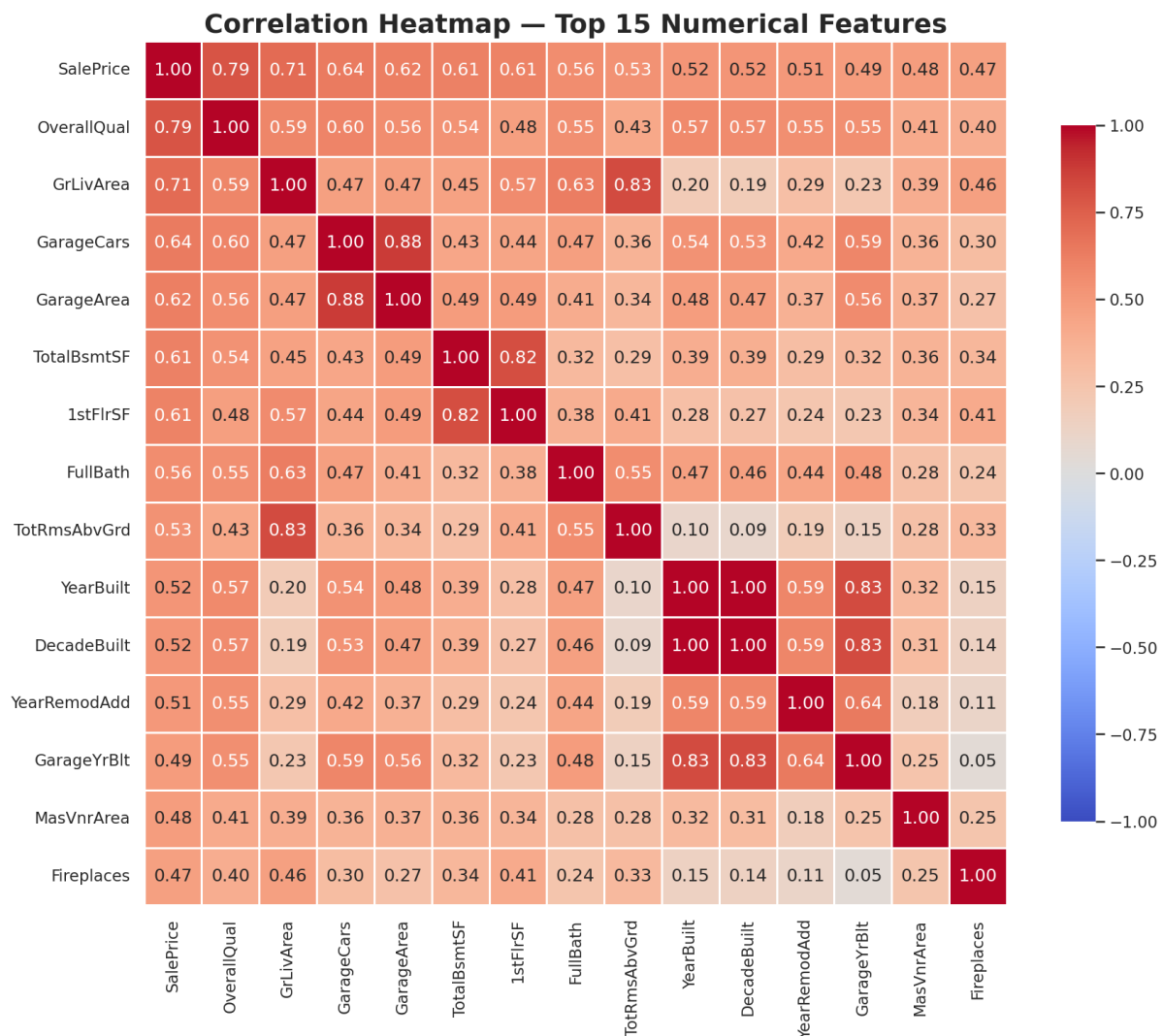


Figure 2: Correlation Heatmap of Top Numerical Features

3.2.1 Key Findings

Strong positive correlations with **SalePrice** included:

- OverallQual (0.79)
- GrLivArea (0.71)
- GarageCars (0.64)
- GarageArea (0.62)
- TotalBsmtSF (0.61)

The heatmap also revealed multicollinearity among several variables:

- YearBuilt and DecadeBuilt
- GarageCars and GarageArea
- TotalBsmtSF and 1stFlrSF

These relationships were later addressed during feature selection.

3.3. Pair Plot Analysis

A pair plot was generated to visualize relationships among major housing features.

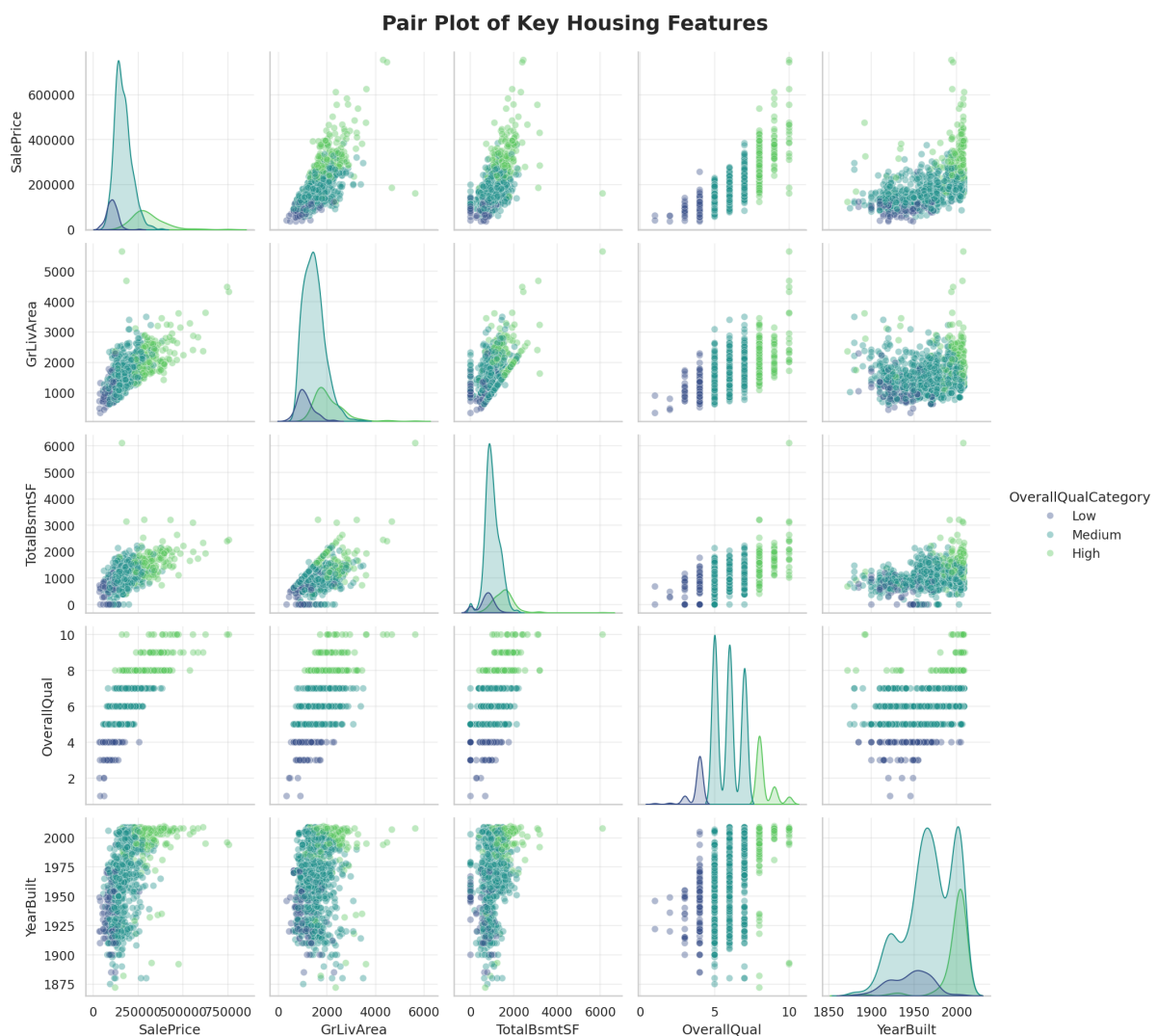


Figure 3: Pair Plot of Key Housing Features

3.3.1 Insights

- Larger living area strongly increased house price

- Higher overall quality corresponded with higher prices
- Newer homes generally had higher quality ratings
- Strong clustering patterns existed among premium homes

3.4. Kitchen Quality Distribution Analysis

Kitchen quality categories were analyzed separately to study their impact on housing prices.



Figure 4: SalePrice Distribution by Kitchen Quality

3.4.1 Observations

- Houses with Excellent (Ex) kitchen quality had the highest average prices
- Typical/Average (TA) kitchens dominated the dataset
- Kitchen quality was strongly associated with property value

3.5. GrLivArea Distribution and Outlier Analysis

Boxplots and violin plots were used to examine the distribution of above-ground living area.

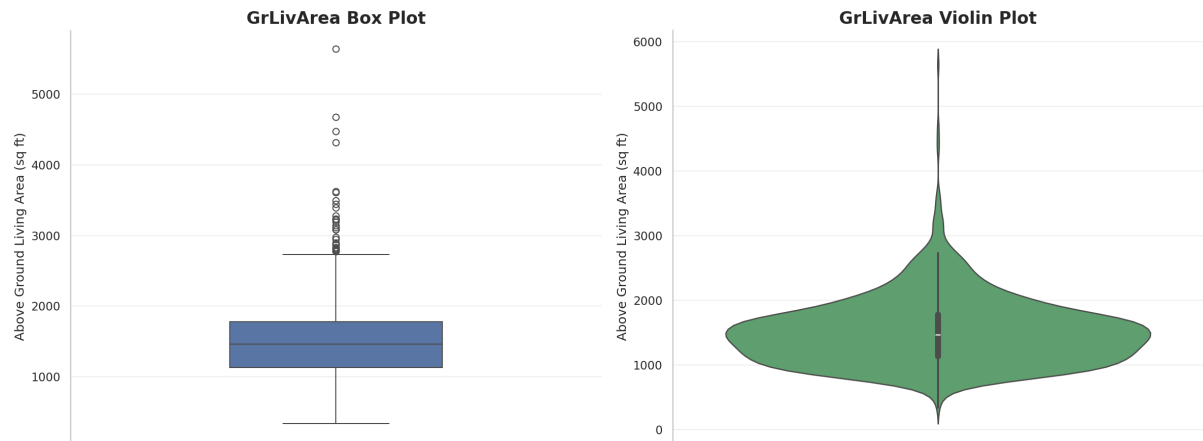


Figure 5: GrLivArea Distribution Analysis

3.5.1 Findings

- Distribution was positively skewed
- Several extreme outliers existed above 4000 square feet
- Most homes clustered between 1000 and 2000 square feet

3.6. Time Trend Analysis

Construction decade trends were analyzed to study market evolution.

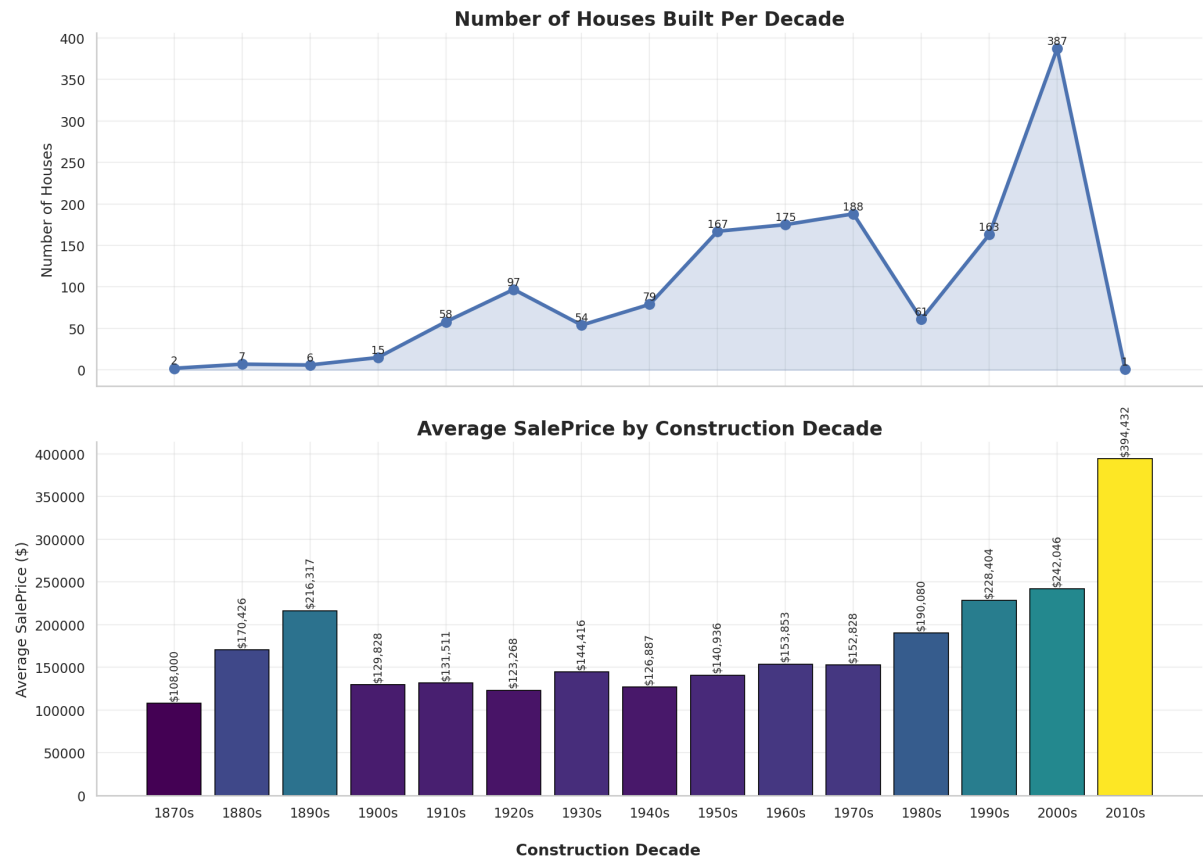


Figure 6: Construction Decade Trends

3.6.1 Insights

- Most homes were built during the 2000s
- Average house prices increased significantly for newer homes
- The 2010s recorded the highest average sale prices

This indicates strong market preference toward modern properties.

4. Feature Engineering

Feature engineering was conducted to improve predictive capability and create more meaningful variables.

4.1. Engineered Features

The following features were created:

Feature	Description
TotalSF	Total basement + first floor + second floor area
TotalBaths	Weighted bathroom count
HouseAge	Age of the house
RemodelAge	Years since remodeling
HasRemodeled	Binary remodeling indicator
QualCond	Combined quality and condition metric
PricePerSF	SalePrice divided by TotalSF
IsNewHouse	House built within the last 5 years

Table 1: Engineered Features

These features improved representation of:

- Property quality
- Home age
- Space utilization
- Market value density

5. Encoding Techniques

Several encoding strategies were used for categorical variables.

5.1. Label Encoding

Ordinal variables such as:

- ExterQual
- KitchenQual
- BsmtQual
- GarageQual

were encoded using ordinal mappings:

$$Ex > Gd > TA > Fa > Po$$

5.2. Frequency Encoding

High-cardinality categorical variables such as:

- Neighborhood
- Exterior1st
- Exterior2nd

were encoded using frequency encoding.

5.3. One-Hot Encoding

Nominal variables without ordering relationships were transformed using one-hot encoding.

Examples:

- MSZoning
- Foundation
- GarageType
- SaleCondition

6. Scaling Analysis

Different feature scaling techniques were compared.

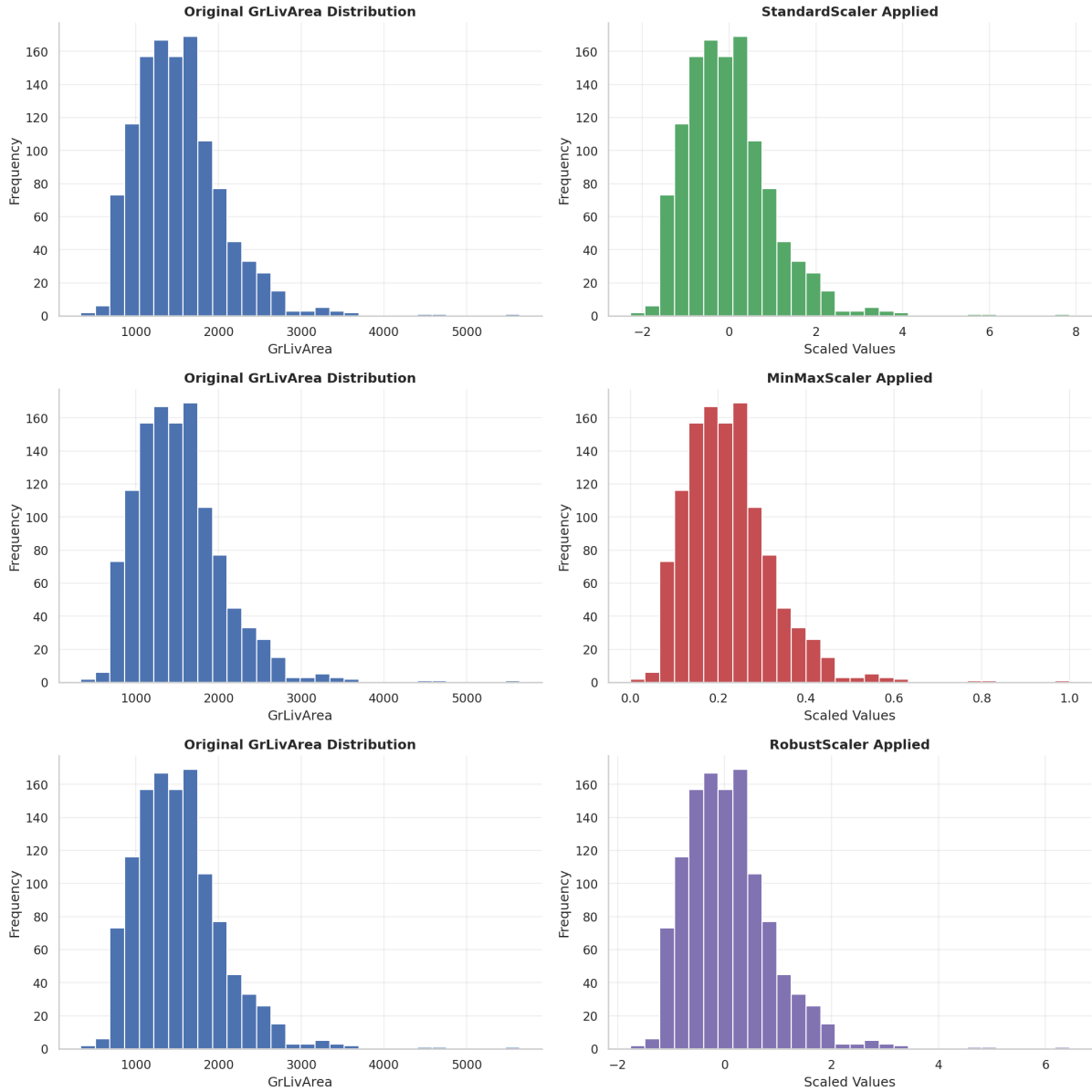


Figure 7: Comparison of Feature Scaling Techniques

6.1. Methods Compared

- StandardScaler
- MinMaxScaler
- RobustScaler

6.2. Findings

- StandardScaler centered data around mean = 0
- MinMaxScaler compressed features into $[0,1]$

- RobustScaler handled outliers more effectively

StandardScaler was selected for the preprocessing pipeline.

7. Skewness Analysis and Transformations

Skewness analysis was performed to identify non-normal features.

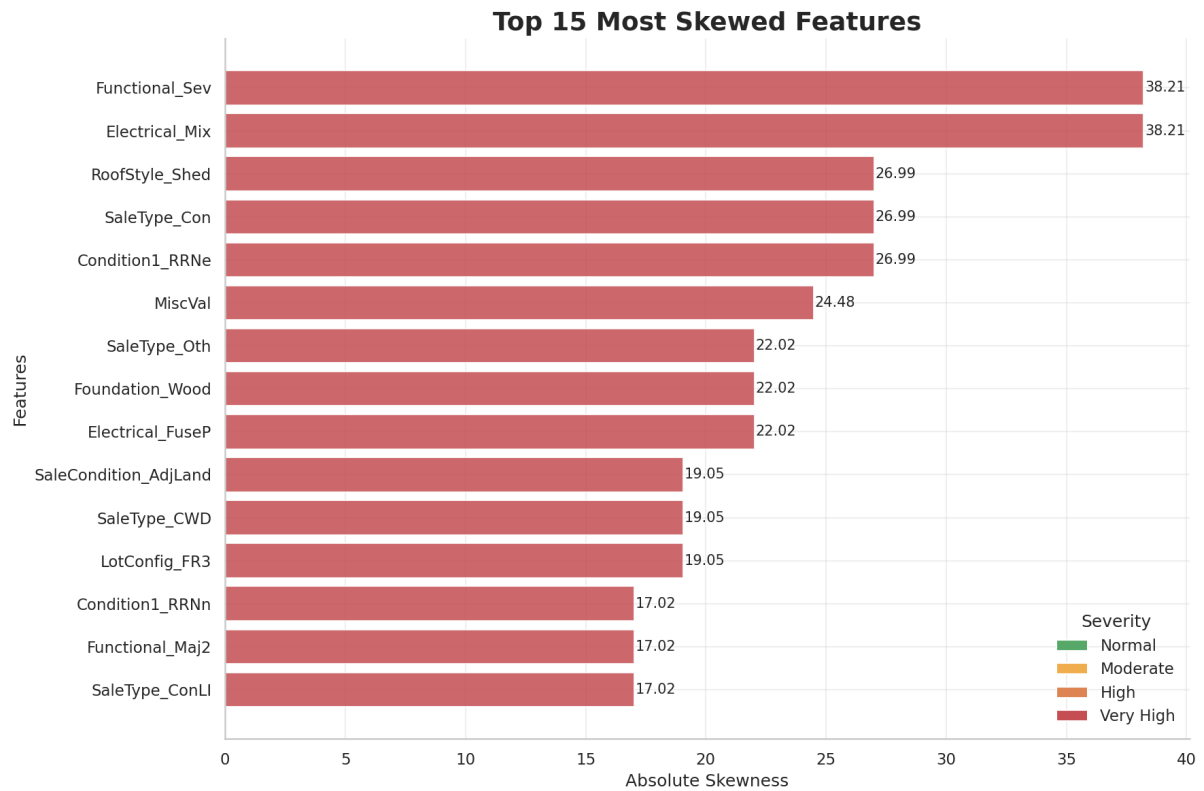


Figure 8: Top 15 Most Skewed Features

7.1. Observations

- 107 features had skewness greater than 1
- Highly skewed variables included:
 - MiscVal
 - LotArea
 - 3SsnPorch

7.2. Transformation Comparison

Different transformations were evaluated.

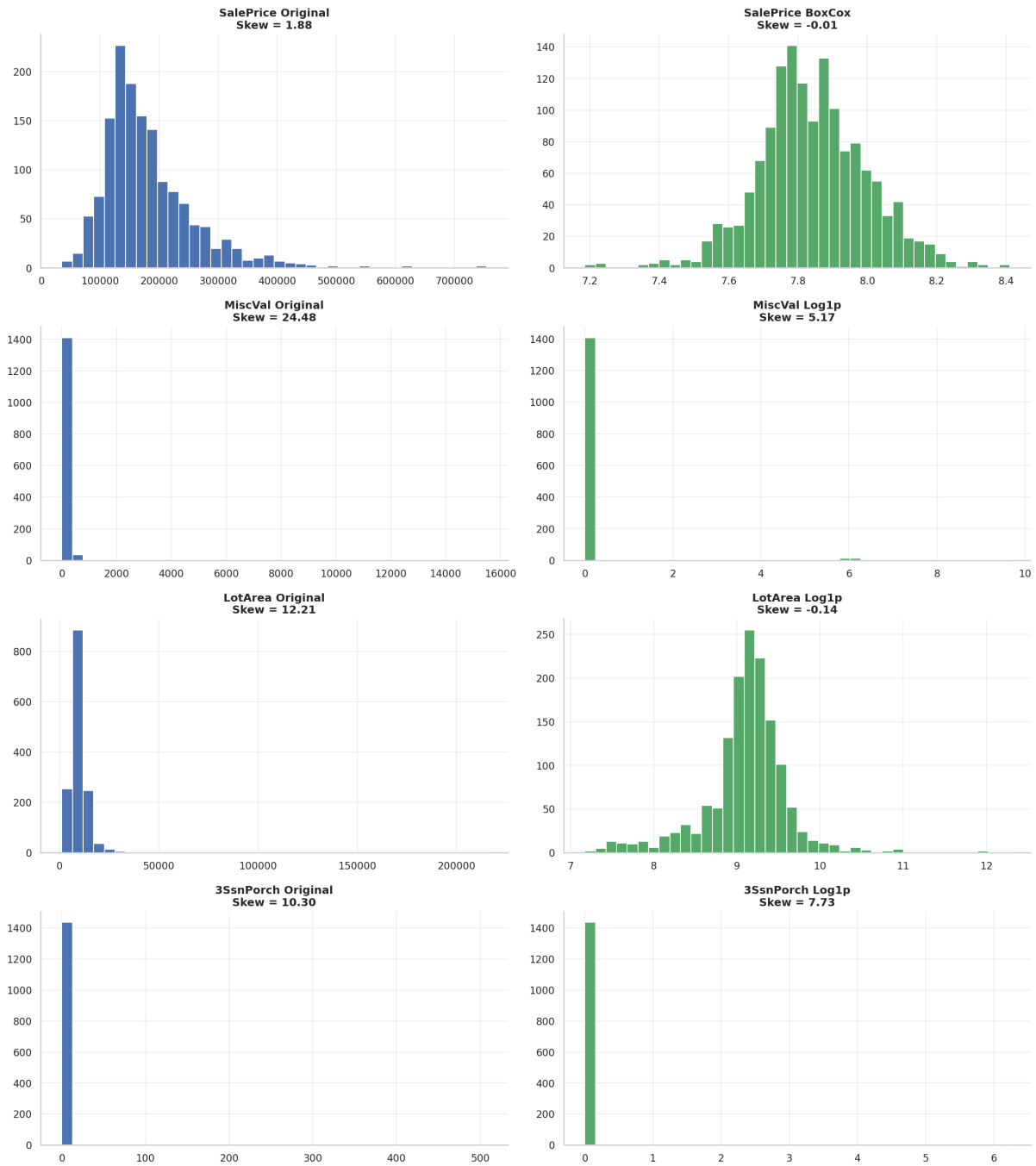


Figure 9: Skewness Reduction Using Transformations

7.3. Results

- Log transformation reduced skewness substantially
- Box-Cox transformation performed best for SalePrice
- Variance stabilization improved model readiness

8. Feature Selection

Feature selection was performed using:

- Correlation filtering
- Variance thresholding
- Multicollinearity removal

8.1. Final Selected Features

Top selected features included:

- OverallQual
- TotalSF
- GrLivArea
- GarageCars
- KitchenQual

8.2. Feature Reduction Summary

- Original Features: 152
- Final Selected Features: 25
- Features Eliminated: 126
- Reduction Percentage: 82.89%

9. Professional Dashboard

A professional dashboard was developed integrating multiple analyses.

House Price Analysis — Professional Dashboard

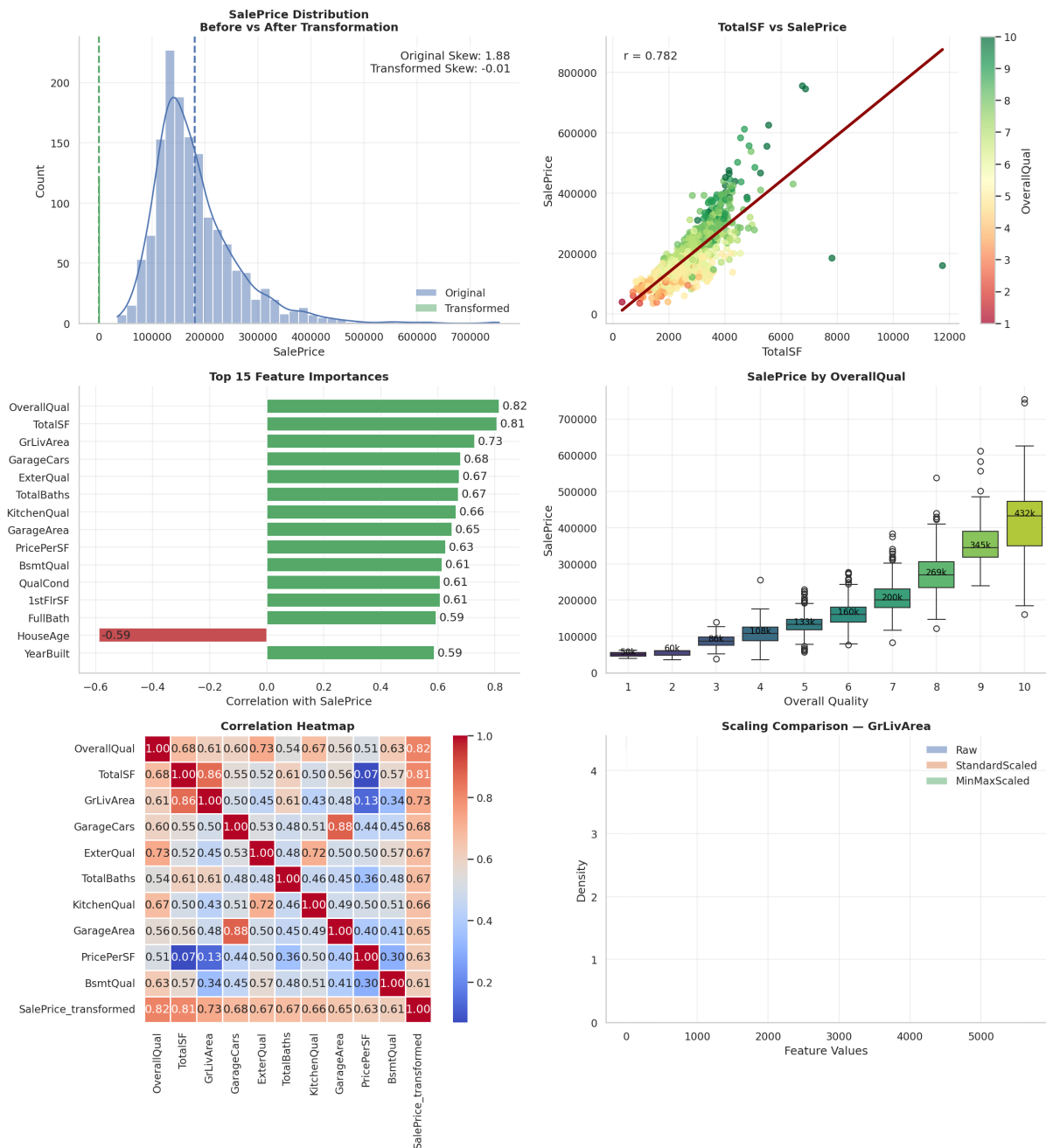


Figure 10: Professional Housing Analysis Dashboard

9.1. Dashboard Components

- SalePrice transformation analysis
- Scatterplot relationships
- Feature importance analysis
- Correlation heatmap

- Scaling comparison

The dashboard provides a complete visual summary of the preprocessing pipeline.

10. Feature Engineering Pipeline Infographic

A summary infographic was created to visualize the complete preprocessing workflow.

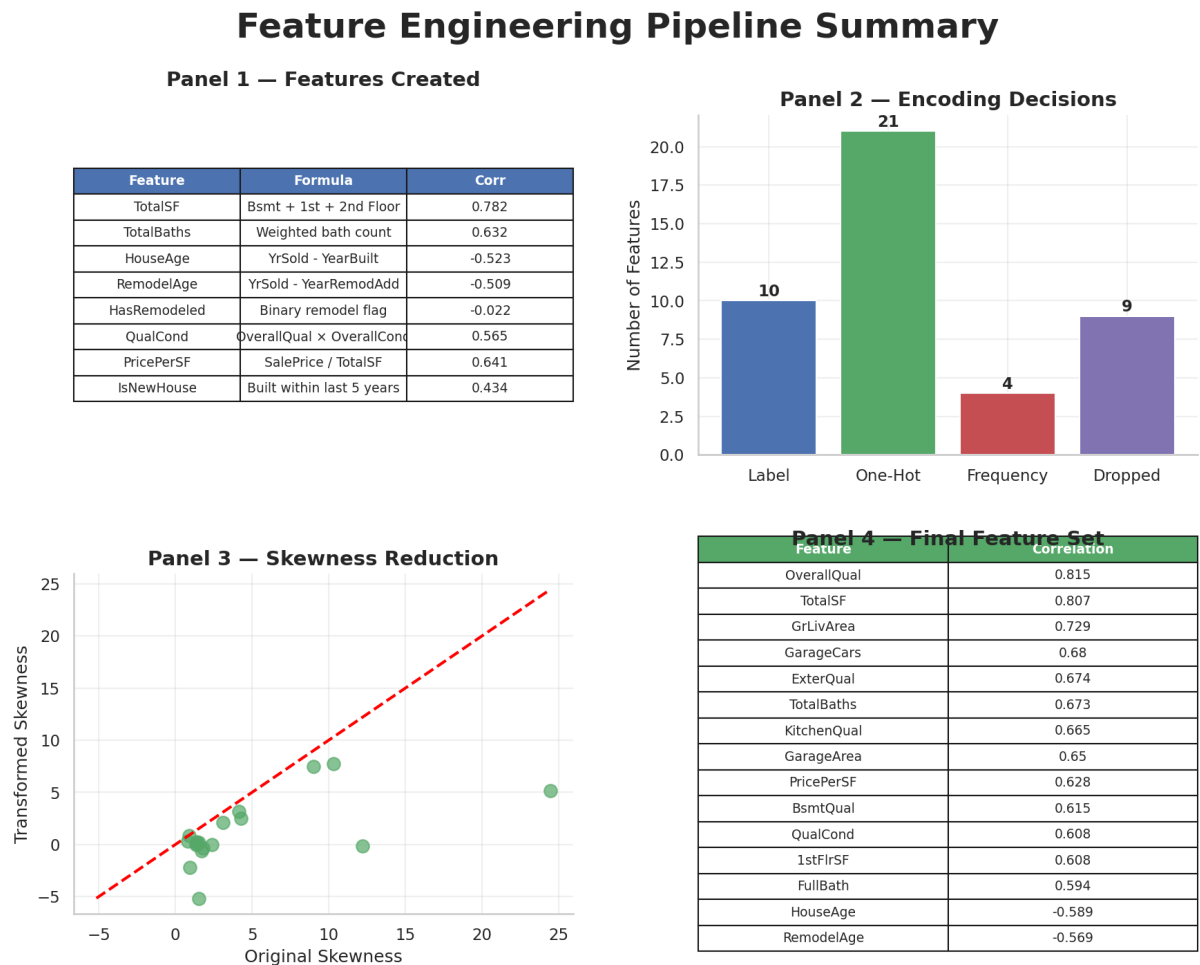


Figure 11: Feature Engineering Pipeline Summary

10.1. Pipeline Highlights

- 8 engineered features created
- Multiple encoding methods applied
- Skewness successfully reduced
- Redundant variables removed
- Optimized feature set prepared

11. Conclusion

This assignment successfully demonstrated a complete exploratory data analysis and feature engineering workflow for a real-world housing dataset.

Major accomplishments included:

- Identifying key predictive variables
- Engineering meaningful features
- Handling skewed distributions
- Encoding categorical variables
- Reducing dimensionality
- Building professional visual dashboards

The final dataset was effectively prepared for machine learning model development through:

- improved feature quality
- reduced multicollinearity
- stabilized distributions
- optimized feature representation

This workflow reflects industry-standard preprocessing practices used in predictive analytics and machine learning pipelines.